Assignment #3

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Overview:

In this Assignment you will create Go programs to process data pipelines. You will execute several experiments to collect measurements and provide comparative analysis for the experimental results. You will create Go programs that utilize concurrency constructs to build batch processing and stream processing for data pipelines.

Submission:

Submit your exercise as a SINGLE ZIP file on Canvas by the due date.
Your submitted ZIP file must have the name:
Assignment_3_Your_LastName.zip

Deliverables:

Your ZIP file for the Assignment submission must include the following:

- All source code that you wrote, installed, compiled, and built on your personal development computer.
- Panopto video recording of a live run (5 to 10 minutes) of your code on your personal development computer.

<u>Requirements Specification:</u>

Implement the following requirements, and test-drive your Go application:

1. Modify and run the **fig-functional-pipeline-combination.go** program to run 2 experiments for every data sample in the following list: 10,000, 100,000, and 1 million random values, and print the time it took to complete the run of every experiment, use the following values for the input of the 2 experiments for every sample:

    1.1. (6 Points) Array/slice of random ints.
    1.2. (6 Points) Array/slice of random float.

2. Modify and run the **fig-pipelines-func-stream-processing.go** program to run 2 experiments for every data sample in the following list: 10,000, 100,000, and 1 million random values, and print the time it took to complete the run of every experiment, use the following values for the input of the 2 experiments for every sample:

    2.1. (6 Points) Array/slice of random ints.
    2.2. (6 Points) Array/slice of random float.

3. Modify and run the **fig-pipelines-chan-stream-processing.go** program to run 2 experiments for every data sample in the following list: 10,000, 100,000, and 1 million random values, and print the time it took to complete the run of every experiment, use the following values for the input of the 2 experiments for every sample:

    3.1. (6 Points) Array/slice of random ints.
    3.2. (6 Points) Array/slice of random float.

4. (12 Points) Modify and run the **fig-pipelines-func-stream-processing.go** program to process the provided Traffic Crashes dataset and print the Zip-Code where the traffic crash occurred

5. (12 Points) Modify and run the **fig-pipelines-chan-stream-processing.go** program to process the provided Traffic Crashes dataset and print the Zip-Code where the traffic crash occurred